

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 5+ years building production systems using TypeScript, Node.js, React, and Express. Led the zero-downtime migration of 20+ applications from Parse/MongoDB to Firebase/GCP at MTailor, cutting \$5,000/month in infrastructure costs while maintaining live production traffic. Expert in real-time data synchronization, high-load API design, and database optimization. Proven track record delivering complex backend systems and responsive React frontends for US companies remotely. Skilled in AWS, Docker, and microservices architecture with Kafka and Kubernetes.

CORE COMPETENCIES

TypeScript / Node.js

React / React Native

Express REST APIs

PostgreSQL / MongoDB / Firestore

Zero-Downtime Migrations

High-Load Systems

AWS / GCP / Firebase

Docker / Kubernetes

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer — US Remote

- **Led end-to-end migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO
- **Architected real-time sync pipeline** using Node.js/TypeScript and Google Pub/Sub for two-way MongoDB-Firestore synchronization during parallel production traffic
- **Built 3D visualization feature** with video overlay using ffmpeg library and React frontend, increasing buyer conversion
- **Reduced infrastructure costs by \$5,000/month** by migrating all services from AWS to GCP
- **Implemented Express Shipping feature** generating additional \$40 revenue per order
- Migrated WebFlow website backend from Parse to Firebase; wrote Python scripts for S3-to-GCS file migration

MTailor

Mar 2022 – Jul 2022

Contractor — US Remote

- **Built migration scripts** in Node.js/TypeScript to read MongoDB data and write to Firestore with complex processing logic
- Authored technical documentation for Firebase SDK integration (Node.js and Python)
- Trained operations team on Firebase Dashboard and backup procedures

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer — US Remote

- **Built DeFi application backends** using Web3 and Node.js/TypeScript for blockchain integrations
- **Delivered React UIs** from Figma designs under tight deadlines, reporting directly to CTO

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- **Optimized PWA performance** delivering real-time stock market prices from Firebase backend
- **Built batch jobs** querying 2+ million Firestore records to sync customer prices from Morningstar APIs
- **Extended Express.js REST APIs** via Firebase Cloud Functions hosted on Heroku
- Added responsive UI pages using React, Tailwind CSS, and HTML

CodeBits

Jan 2020 – Jul 2021

Software Engineer — Uganda Remote

- **Led team of 4 developers** building legal tech systems for NGOs (FIDA Uganda, LASPNET Uganda)
- **Architected microservices backend** using Apache Kafka, Docker, and Kubernetes for case management app
- Built cross-platform mobile app with React Native/Expo (iOS & Android) with Firebase push notifications
- Developed USSD service for legal aid providers communicating with Node.js backend via gRPC

KEY PROJECTS

Parse → Firebase Migration (MTailor) Production

Led zero-downtime migration of 20+ applications with live Pub/Sub synchronization pipeline. Saved \$5K/month in infrastructure costs.
Node.js, TypeScript, Firebase, GCP Pub/Sub, MongoDB, Firestore

Real-time Stock Price PWA (Dr Wealth) Production

Optimized PWA querying 2M+ Firestore records for real-time Morningstar price data.
React, Express.js, Firebase Cloud Functions, Firestore

Legal Case Management System (FIDA Uganda) Production

Microservices architecture with Kafka message streaming for NGO case management.
Node.js, Apache Kafka, Docker, Kubernetes, React Native

EDUCATION

Bachelor of Science in Computer Engineering — **Makerere University**

2018 – 2022

Kampala, Uganda

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Go (learning) **Backend:** Node.js, Express, Firebase Cloud Functions, gRPC

Frontend: React, React Native, Tailwind CSS, HTML/CSS **Databases:** PostgreSQL, MongoDB, Firestore, Redis

Cloud: GCP (Pub/Sub, Cloud Functions), AWS (S3, EC2, EBS), Firebase **DevOps:** Docker, Kubernetes, Git, Linux

Architecture: Microservices, REST APIs, Event-driven (Kafka, Pub/Sub)